Amir Hajian, Ph.D.

AI Research & Product Exec — multimodal agentic AI, AI for Science, post-training and research-to-product leadership

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Seasoned AI research executive and scientist with 10+ years building and leading multi-disciplinary teams of researchers, engineers, and data specialists across academia and industry. Deep expertise in multimodal and generative AI, responsible AI (robustness, bias, safety, evaluation), and post-training adaptation of foundation models in enterprise and regulated contexts. Author of 70+ peer-reviewed publications (h-index 47); strong track record translating frontier research into scalable, trustworthy production systems—from wearables to agentic AI in financial services. Experienced in organizational planning, mentorship, and growing senior technical talent.

CURRENT

Head of Arteria Research Lab - Arteria AI

Toronto, ON — 2024 - Present

Built and currently heading up Arteria AI Cafe, a research lab for innovating novel techniques in multimodal agentic AI (NLP and computer vision) powered by GenAI. Leading science, engineering, MLOps and product teams to take cutting edge research to product.

- Lead the design and building of DAX: a fully agentic suite of applications for all documentation tasks. 40x improvement to efficiency compared to the existing methods. Transformed how Arteria solved client problems.
- VLM/LLM post-training from research to production.
- Full stack agentic AI app development: from infrastructure to launch in production.
- Building powerful multimodal GenAI products based on in-house innovative research on agentic AI that learns in real time.
- Leading research in cutting edge AI, with publications in major conferences including AAAI 2024 and The ACM WebConf 2025.
- Successfully merged cutting edge research with efficient product development to go from ideation to real tools in a short time.
- Tested the resulting products with clients in a highly competitive market and used a modern iterative process to launch successful products that users loved.

VP of AI & Chief Scientist - Arteria AI

Toronto, ON — 2021 - 2024

Built and managed the R&D organization consisting of teams of AI

EDUCATION

Princeton University
Postdoctoral Researcher, Data Driven Cosmology

Inter University Centre for Astronomy and Astrophysics
***Ph.D.* Astrophysics**

Institute for Advanced Studies in Basic Sciences
Master of Science, Physics.

Sharif University of Technology
Bachelor of Science, Physics.

researchers, data scientists, ML Engineers, MLOps and annotators that I built and trained from scratch. That led to Arteria being named as the Most Promising 100 AI Startups by CB Insights and one of the 3 "Coolest GenAI startups in FinTech" By Gartner in 2023.

- Led the end-to-end design and building of CUCINA: an enterprise-grade, compliance-ready multimodal document understanding system for regulated financial services, extracting structured information from the most complex document layouts using Foundation Models.
- Invented a novel multimodal architecture advancing the state of the art in vision-language modeling, giving Arteria a decisive edge. Published a paper on the method and architecture, presented in AAAI 2024.
- Making use of Large Language Models (LLMs) and Generative AI for efficient information extraction
- Led Arteria to win large contracts in AI bake offs with multiple clients
- Founded MLOps at Arteria for efficient productionization of large, complex deep learning models. Automated an optimal CI/CD pipeline for using large language models in production.
- Built the most efficient AI model training and serving possible today
- Built a highly successful AI team and designed a workflow that allowed for quick turnarounds of ideas to be developed fast to go to production.
- Fostering an inclusive organization culture that promotes creativity, collaboration, transparency, and continuous innovation.
- Led the design and implementation of a highly efficient infra-structure required for AI product and high performance computing development and deployment in the cloud.
- Mentored UWaterloo MS students on Multimodal and Computer Vision research, guiding them from academic thesis to career-track applied scientists contributing directly to production model development.
- Presented in multiple seminars and meetups about our work on LLMs, multimodal models and MLOps in the age of Transformers.

EXPERIENCE

Director of Applied Research - Scribd

Toronto, ON — 2019 - 2021

Founded Scribd's Applied Research Team in Toronto to build products using machine learning to help customers explore the library, discover great content and read what they love. Designed the research & product roadmap and built and led a strong team to deliver on that.

- Hands on AI product management: led the company's efforts to create innovative AI-products using novel machine learning

SKILLS

Leadership: Building and managing high performing, happy and diverse teams on the foundations of innovation, inclusivity, constant learning and collaborative team culture.

Product: Building powerful products

methods for search and recommendation (blog posts here, here and here).
- Worked with senior leadership to design roadmaps and defined directions and goals in AI research and to take results to production.
- Built end to end products, from ideation to production, leading distributed teams of researchers, engineers and project managers
- Created a superior user experience using the state of the art in computer vision, NLP, deep learning, statistical modeling and distributed computing in the cloud. The product is the backbone of the company's content understanding today.
- Fostering a team culture that promotes creativity, collaboration, transparency, and innovation.
- Led the design and implementation of the infra-structure required for AI product development in the cloud.
- Taught a hands-on workshop on Deep Learning using PyTorch for computer vision.

Director of AI Research, Thomson Reuters Labs
Toronto, ON — 2017 - 2019

Led company's efforts to build AI-powered solutions at scale to create value and to help the company discover, explore and de-risk novel ways of generating revenue inspired by the state of the art technology and shorten time to market:

- Designed and led the end-to-end development of an AI as a Service product in the cloud for creating a searchable video database using computer vision, deep learning, cloud computing and NLP in an agile setting. Product Launched in May 2019, and has been used in production since then.
- Led company's initiative in design and development of an end-to-end AI-powered Risk product from scratch with scientists and engineers. Product launched in May 2018. Patent pending. Featured on the company's AI timeline.
- Hiring and leading teams of researchers and engineers to build products using advanced technologies.
- Innovation with partners: connecting with universities and startups to build relationships and collaborations
- Presented in meetups, conferences & taught in universities (Harvard and UofT). See examples on Transitioning to Data Science (here), Machine learning in production (here) and our presentations at Strata (here) and Canadian AI (here).
- Organized Machine Learning for Finance Professionals Workshop in collaboration with CFA Society, Toronto
- Organizer of Toronto Deep Learning Meetup since 2015.

Senior Data Scientist, Thomson Reuters Labs
Toronto & Waterloo, ON — 2015 - 2017

Founded the labs in Canada. Actively engaged in building the network for the labs with the startup community in Toronto and Waterloo and the customers worldwide. SME and point of contact for data-driven product development in relationship with the business units and customers. Built new solutions by rapid prototyping and iterative customer validation.

- Initiated and built the scientific team in the labs

using deep multimodal models combining computer vision and NLP.

Computer Vision: Vision Language Models (VLMs), Diffusion and Flow Matching GenAI, Object detection, text extraction, scene extraction and semantic segmentation in videos and images.

NLP: using the best of the Transformers architecture to build high impact products. LLM based Language agents for solving complex problems

Responsible AI & Alignment: Post-training and fine-tuning of foundation models; evaluation frameworks and benchmarks; robustness, bias/fairness analysis, safety, and explainability in enterprise and regulated contexts

MLOps: Optimizing models and pipelines to use massive deep learning models in production at the lowest cost, least latency and shortest development time

Programming: Python, PyTorch, JAX, Scala; cloud infrastructure on Databricks

SAMPLE PRODUCTS

- Fully Automated Multimodal Document Understanding with Transformers-based deep NLP and computer vision
- Generating book covers using generative diffusion and flow models
- Efficient object detection for information extraction in real time
- Reuters Content Access Product: object, people and scene detection in videos using deep transfer learning;
- WorldCheck Product: End-to-end risk mining solution;
- Eikon Product: Scalable

- Built 17 POCs in 18 months
- Successfully led projects from problem definition to production: *Recommendation Engine, White House Run App.*
- Frequent customer visits & academic meetings
- Presented in conferences & taught in universities in Canada and the US

Senior Research Associate, CITA (University of Toronto)
Toronto, ON — 2010 - 2015

Designed Machine Learning algorithms to model the universe. Led large international collaborations to extract information from massive noisy data. Supervised research teams and students. Spearheaded creating a modern scientific computing framework for all computing needs of the team.
Consulted a startup, solved data problems for building wearables devices.

Research Scientist, Department of Physics (Princeton University)
Princeton, NJ — 2006-2010

Designed and built an end to end data analysis pipeline used as the backbone of a large collaborative effort to measure the properties of the universe based on multiwavelength observations using Bayesian statistics and probabilistic programming.

recommender systems;
- Reuters Product: US Election simulator;
- Denoising bio-signal time series for wearable devices;
- Bank distress detection using language patterns in the news;
- Time series & anomaly detection;
- Opinion mining;
- Coreference resolution for intelligent tagging;
- Scalable multi-lingual document similarity

ACADEMICS

70+ peer reviewed articles in top scientific journals (h-index 47). Popular science articles in the newspapers and magazines. Scientific editor for Scala Data Analysis Cookbook. Full list of publications (here).

	All	Since 2020
Citations	9243	2158
h-index	47	27
i10-index	65	46

MY WORK IN THE NEWS